

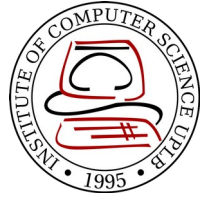
CMSC 137

Data Communications and Networking

Lecture Module
19 – Internet Protocol



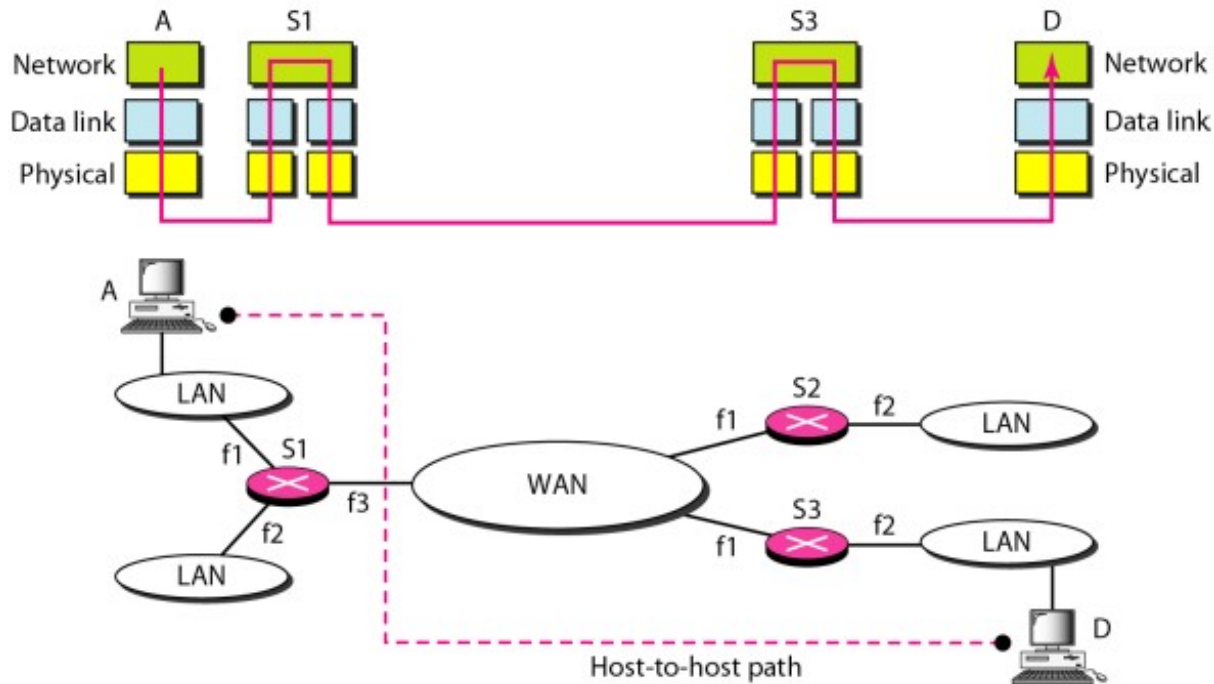
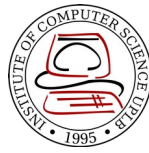
University of the Philippines
LOS BAÑOS



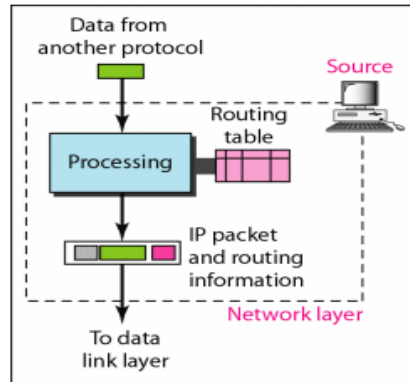
Module 19: Internet Protocol

Rozano S. Maniaol
Lecturer

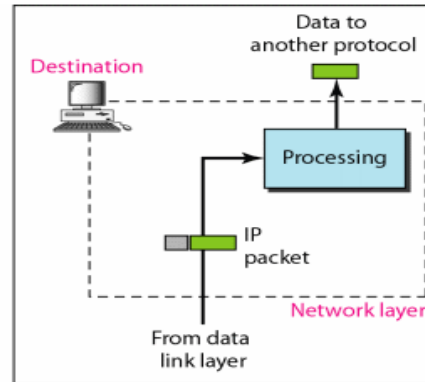
Internetworking



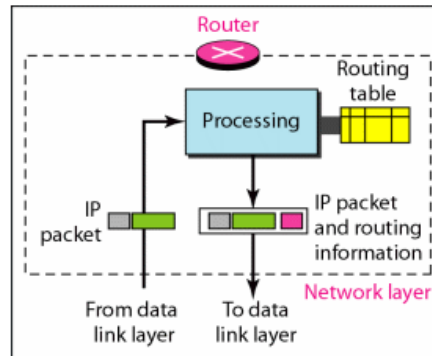
Internetworking



a. Network layer at source

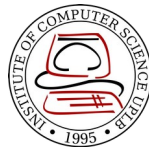


b. Network layer at destination

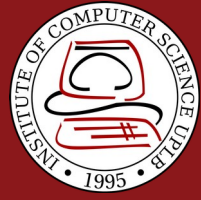


c. Network layer at a router

The Internet at the Network Layer

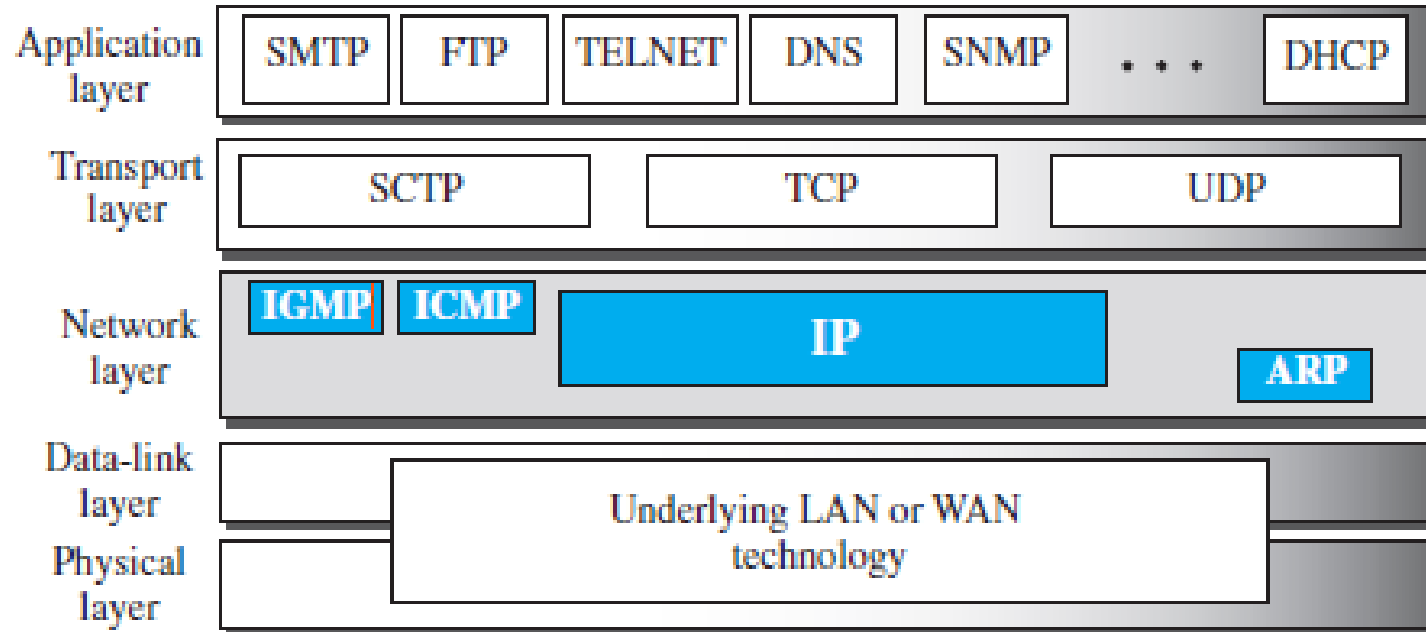
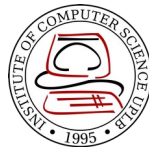


- Uses the datagram approach to packet switching
- Uses the universal address defined in the network layer to route packets from the source to the destination
- Communication is connectionless
- Treats each packet independently

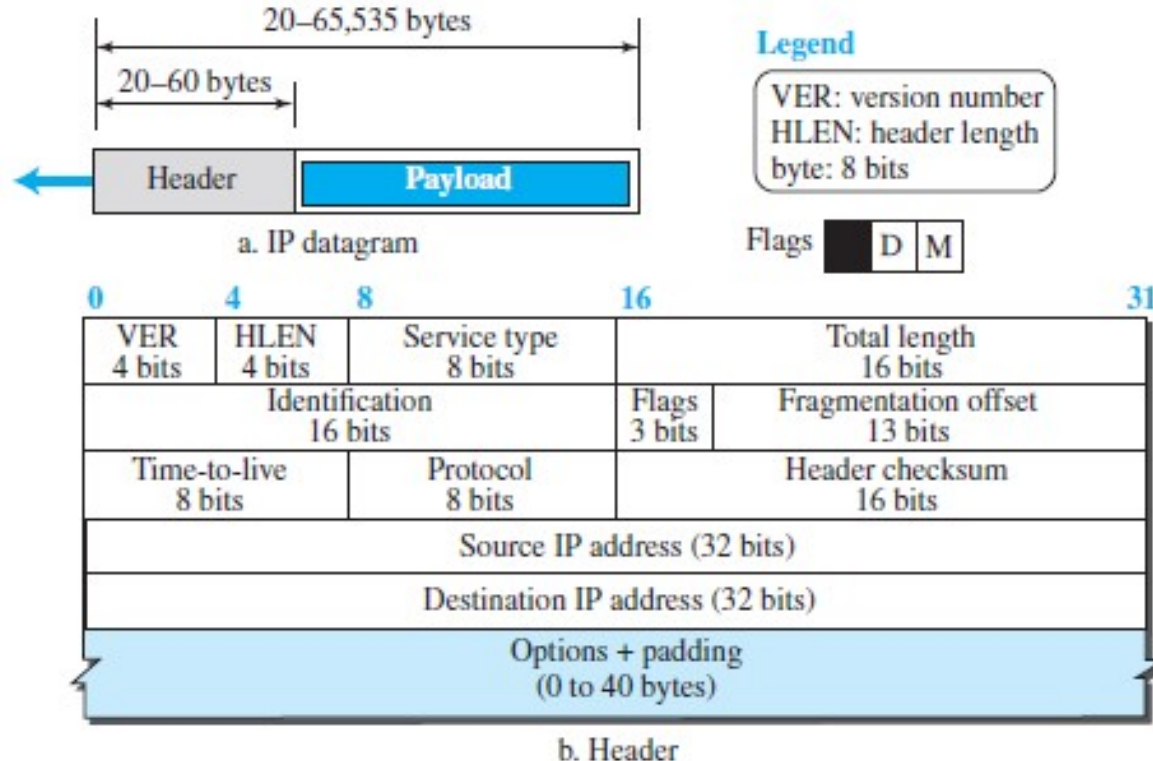


IPv4

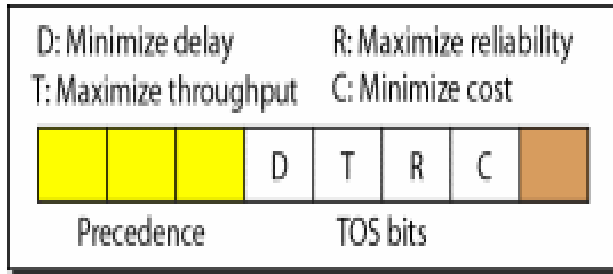
IPv4 in the TCP/IP Protocol Suite



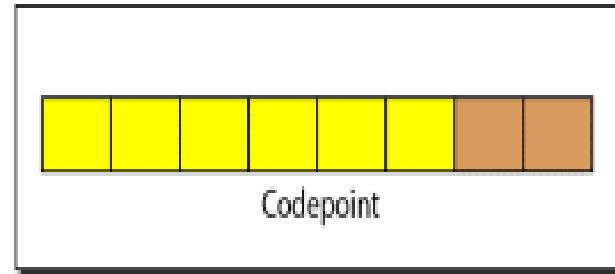
IPv4 Datagram Format



- **Version** (VER) – 4-bit field that defines the version of the IPv4 protocol
- **Header length** (HLEN) – 4-bit field that defines the total length of the datagram header in 4-byte words
- **Service** – previously called service type, now called differentiated service



Service type

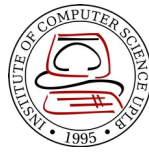


Differentiated services

- **Precedence** - 3-bit subfield that defines the priority of datagram in issues such as congestion
- **TOS** (Type of Service) bits - 4-bit subfield with each bit having a special meaning

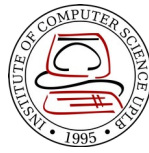
<i>TOS Bits</i>	<i>Description</i>
0000	Normal (default)
0001	Minimize cost
0010	Maximize reliability
0100	Maximize throughput
1000	Minimize delay

Service Type of Some Applications



<i>Protocol</i>	<i>TOS Bits</i>	<i>Description</i>
ICMP	0000	Normal
BOOTP	0000	Normal
NNTP	0001	Minimize cost
IGP	0010	Maximize reliability
SNMP	0010	Maximize reliability
TELNET	1000	Minimize delay
FTP (data)	0100	Maximize throughput
FTP (control)	1000	Minimize delay
TFTP	1000	Minimize delay
SMTP (command)	1000	Minimize delay
SMTP (data)	0100	Maximize throughput
DNS (UDP query)	1000	Minimize delay
DNS (TCP query)	0000	Normal
DNS (zone)	0100	Maximize throughput

Differentiated Services

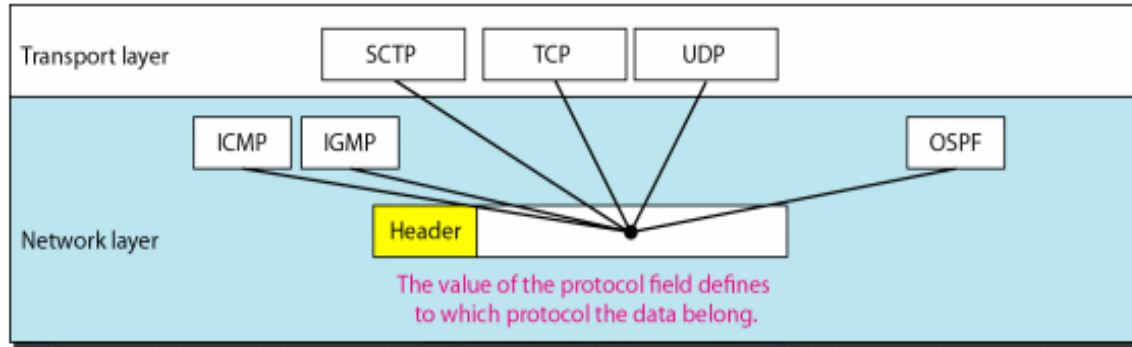


- First 6 bits make up the codepoint subfield, last 2 bits are not used
 - When the 3 rightmost bits are 0s, the 3 leftmost bits are interpreted the same as the precedence bits
 - When the 3 rightmost bits are not all 0s, the 6 bit define 64 services based on the priority assignment by the Internet or local authorities

<i>Category</i>	<i>Codepoint</i>	<i>Assigning Authority</i>
1	XXXXX0	Internet
2	XXXX11	Local
3	XXXX01	Temporary or experimental

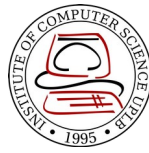
- **Total length** – 16-bit field that defines the total length of the IPv4 datagram in bytes
- **Identification** – used in fragmentation
- **Flags** – used in fragmentation
- **Fragmentation offset** – used in fragmentation
- **Time to live (TTL)** – used to control the maximum number of hops (routers) visited by the datagram

- **Protocol** – 8-bit field that defines the higher-level protocol that uses the services of the IPv4 layer



<i>Value</i>	<i>Protocol</i>
1	ICMP
2	IGMP
6	TCP
17	UDP
89	OSPF

IPv4



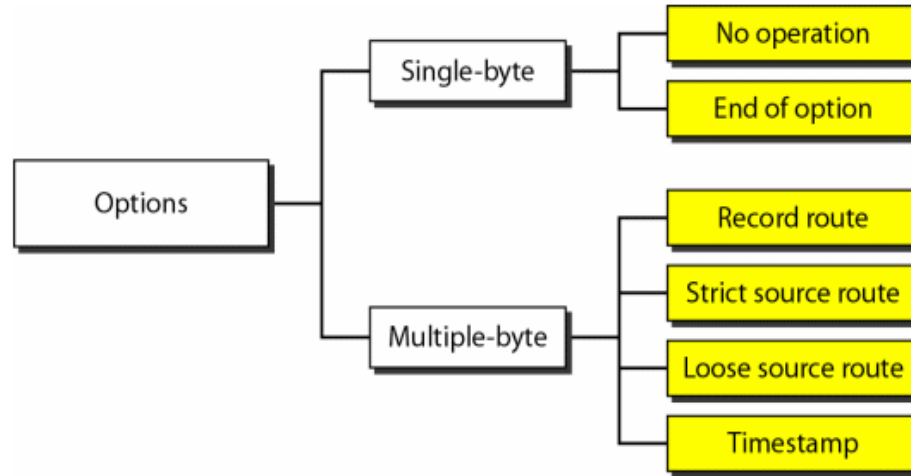
- Checksum – for error checking, covers only the header

4	5	0	28	
49.153			0	0
4	17	0		
10.12.14.5				
12.6.7.9				

4, 5, and 0	→	4	5	0	0
28	→	0	0	1	C
1	→	C	0	0	1
0 and 0	→	0	0	0	0
4 and 17	→	0	4	1	1
0	→	0	0	0	0
10.12	→	0	A	0	C
14.5	→	0	E	0	5
12.6	→	0	C	0	6
7.9	→	0	7	0	9
Sum	→	1	3	4	E
Wrapped sum	→	3	4	4	F
Checksum	→	C	B	B	0

Replaces 0

- **Source Address** – 32-bit address that defines the IPv4 address of the source
- **Destination Address** – 32-bit address that defines the IPv4 address of the destination
- **Options** – maximum of 40 bytes, used for network testing and debugging

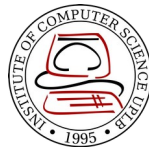


Options



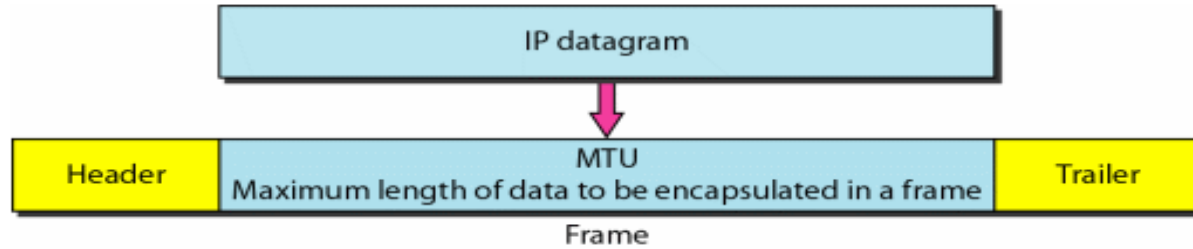
- **No operation** – 1-byte option used as a filler between options
- **End of option** – 1-byte option used for padding at the end of the option field
- **Record route** – used to record the Internet routers that handle the datagram (up to 9 router addresses)
- **Strict source route** – used by the source to predetermine a route for the datagram as it travels through the Internet
- **Loose source route** – similar to strict source route in which each router in the list must be visited, but the datagram can visit other routers as well
- **Timestamp** – used to record the time of datagram processing by a router

Fragmentation



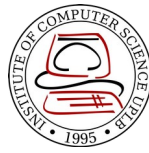
- Each data link protocol has its own frame format in most protocol, where one of the fields defined in the format is the maximum size of the data field, the Maximum Transfer Unit or **Maximum Transmission Unit** (MTU)
- When a datagram is encapsulated in a frame, the total size of the frame must be less than the MTU, which is defined by the restrictions imposed by the hardware and software used

Fragmentation

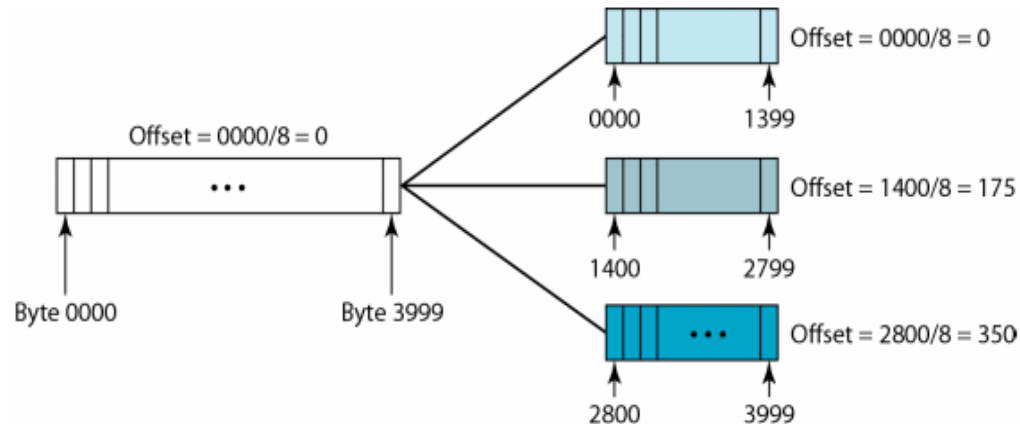


<i>Protocol</i>	<i>MTU</i>
Hyperchannel	65,535
Token Ring (16 Mbps)	17,914
Token Ring (4 Mbps)	4,464
FDDI	4,352
Ethernet	1,500
X.25	576
PPP	296

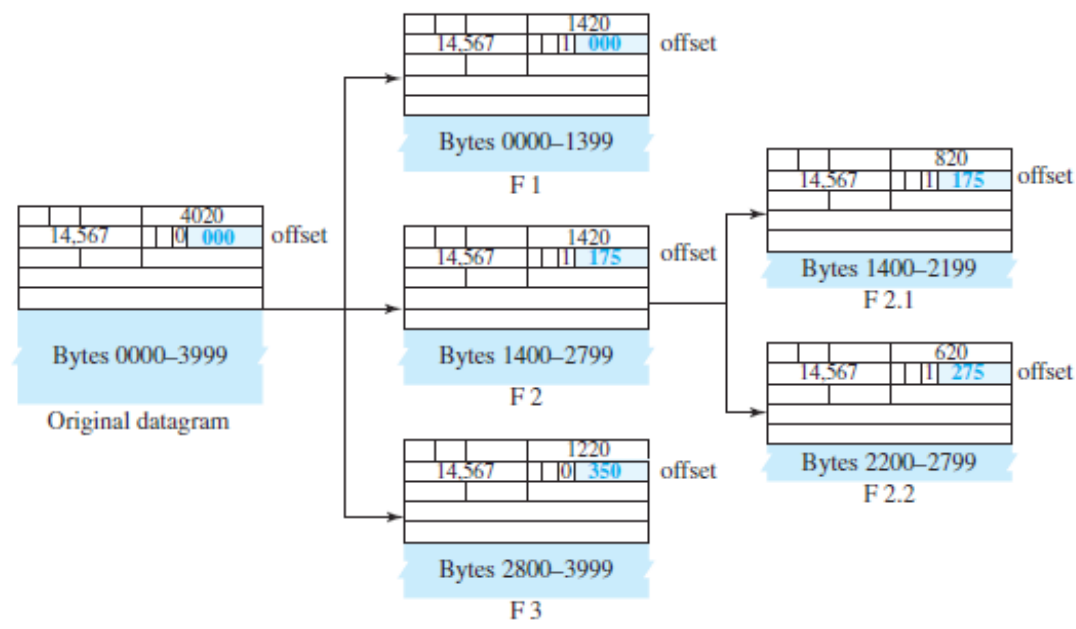
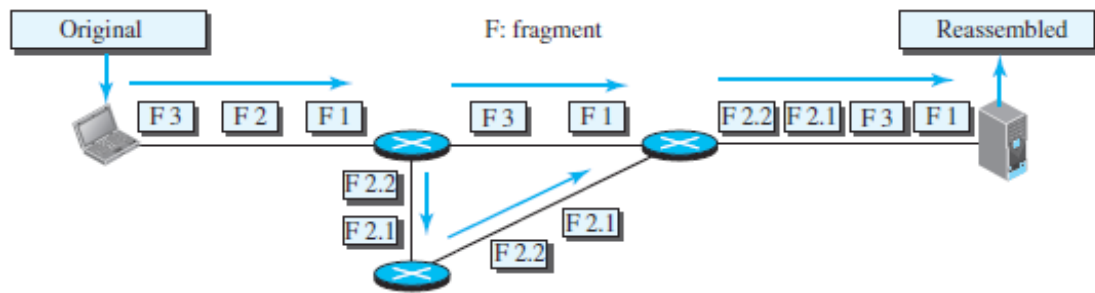
Fragmentation

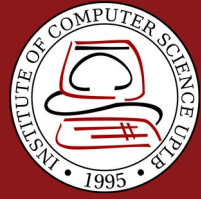


- **Identification** – 16-bit field that identifies the datagram originating from the source host
- **Flags** – 3-bit field; 1st bit is reserved, 2nd bit is called do not fragment bit, 3rd bit is more fragment bit
- **Fragmentation offset** – 13-bit field that shows the relative position of the fragment with respect to the whole datagram



Fragmentation



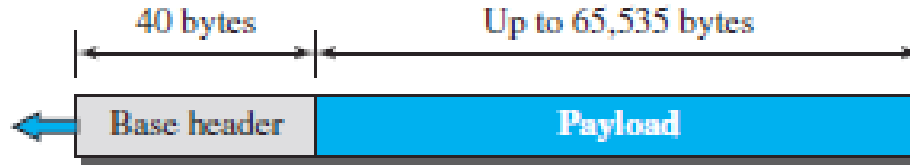
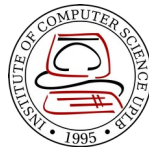


IPv6

- Internetworking Protocol, version 6, also known as **IPng** (Internetworking Protocol, next generation)
- To overcome deficiencies of IPv4
- Address depletion is a long-term problem
- Internet must accommodate real-time audio and video transmission
- Internet must accommodate encryption and authentication of data

- Advantages
 - Larger address space
 - Better header format
 - New options
 - Allowance for extension
 - Support for resource allocation
 - Support for more security

IPv6 Header and Payload

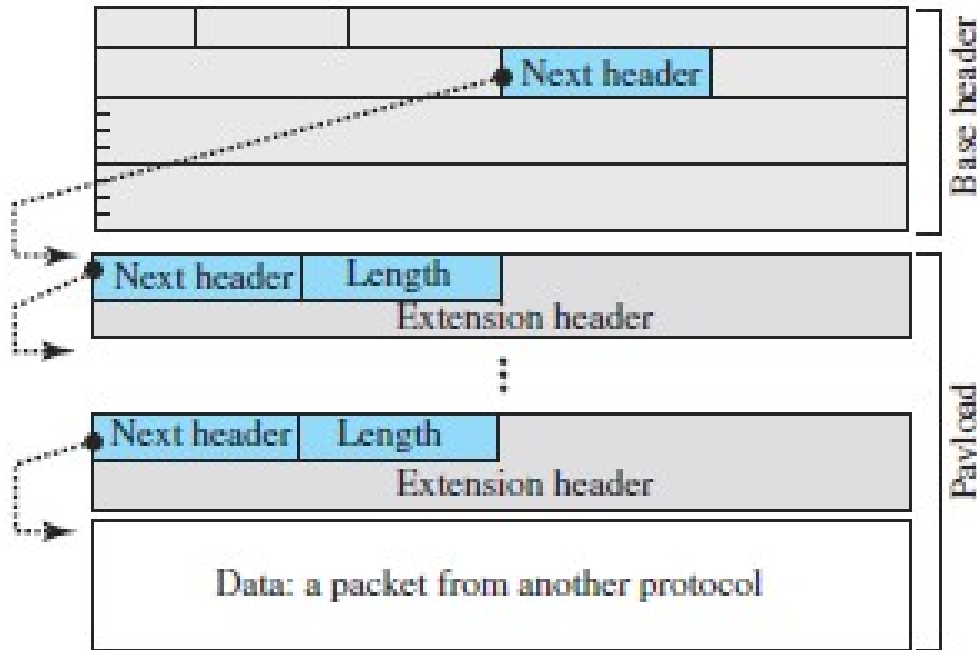
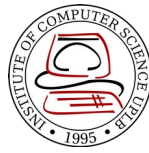


a. IPv6 packet

0	4	12	16	24	31
Version	Traffic class	Flow label			
Payload length			Next header	Hop limit	
Source address (128 bits = 16 bytes)					
Destination address (128 bits = 16 bytes)					

b. Base header

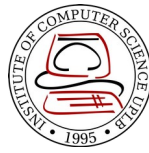
IPv6 Payload



Some next-header codes

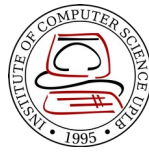
- 00: Hop-by-hop option
- 02: ICMPv6
- 06: TCP
- 17: UDP
- 43: Source-routing option
- 44: Fragmentation option
- 50: Encrypted security payload
- 51: Authentication header
- 59: Null (no next header)
- 60: Destination option

IPv6 Base Header



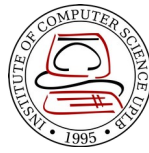
- **Version** – 4-bit field that defines the version number of the IP
- **Traffic class** – 8-bit field that distinguishes different payloads with different delivery requirements
- **Flow label** – 20-bit field that is designed to provide special handling for a particular flow of data
 - A flow is a sequence of packets that share the same characteristics
- **Payload length** – 2-byte field that defines the length of the IP datagram excluding the base header

IPv6 Base Header



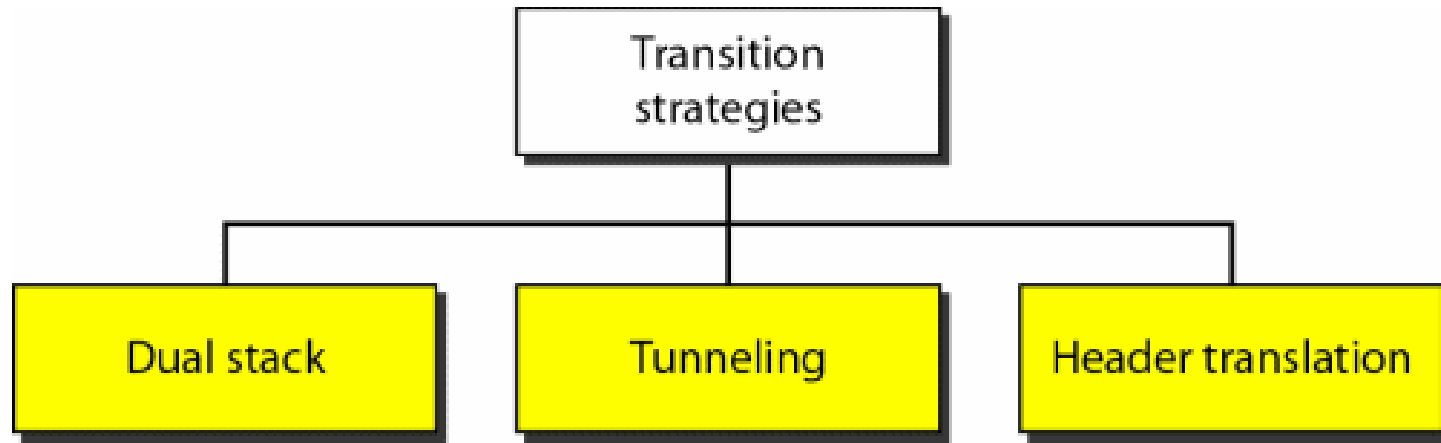
- **Next header** – 8-bit field defining the type of the first extension (if present) or the type of data that follows the base header
- **Hop limit** – 8-bit field that serves the same purpose of TTL field in IPv4
- **Source address** – 16-byte IP address that identifies the original source of the datagram
- **Destination address** – 16-byte IP address that identifies the final destination of the datagram

IPv4 vs IPv6 Header



<i>Comparison</i>	
1.	The header length field is eliminated in IPv6 because the length of the header is fixed in this version.
2.	The service type field is eliminated in IPv6. The priority and flow label fields together take over the function of the service type field.
3.	The total length field is eliminated in IPv6 and replaced by the payload length field.
4.	The identification, flag, and offset fields are eliminated from the base header in IPv6. They are included in the fragmentation extension header.
5.	The TTL field is called hop limit in IPv6.
6.	The protocol field is replaced by the next header field.
7.	The header checksum is eliminated because the checksum is provided by upper-layer protocols; it is therefore not needed at this level.
8.	The option fields in IPv4 are implemented as extension headers in IPv6.

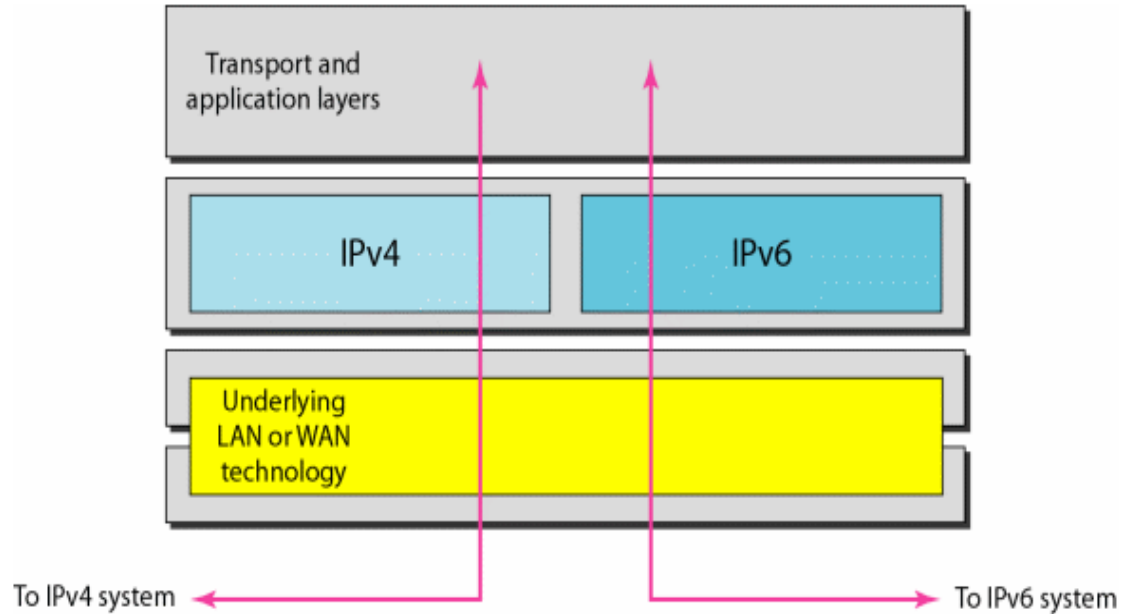
Transition from IPv4 to IPv6



Dual Stack



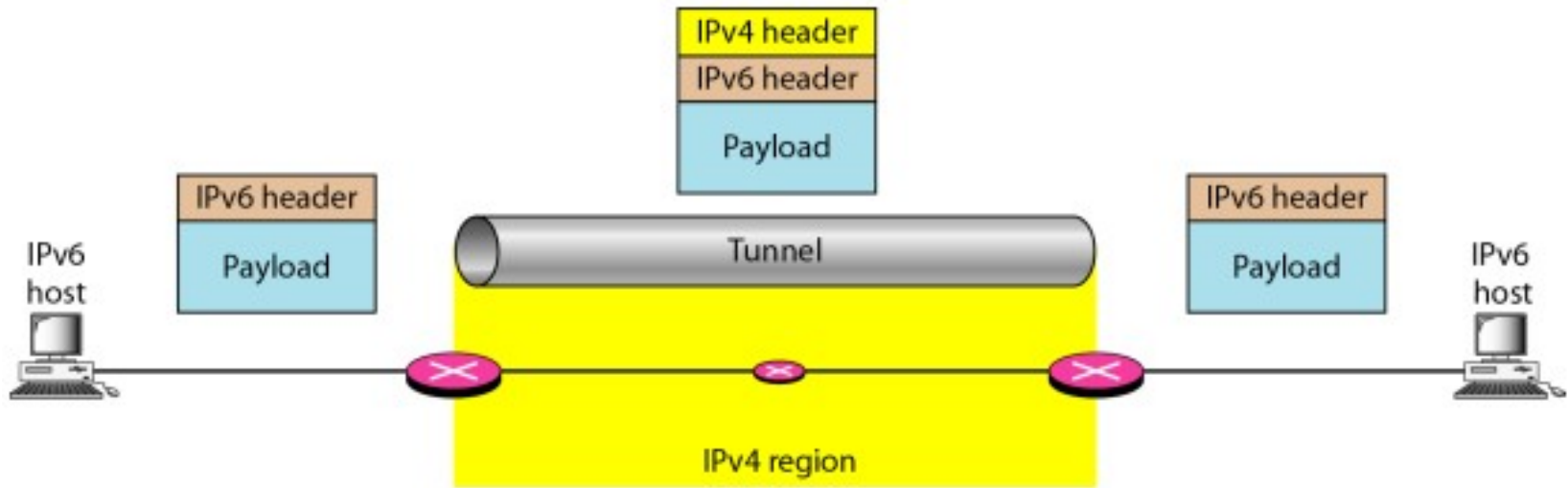
Station must run IPv4 and IPv6 simultaneously



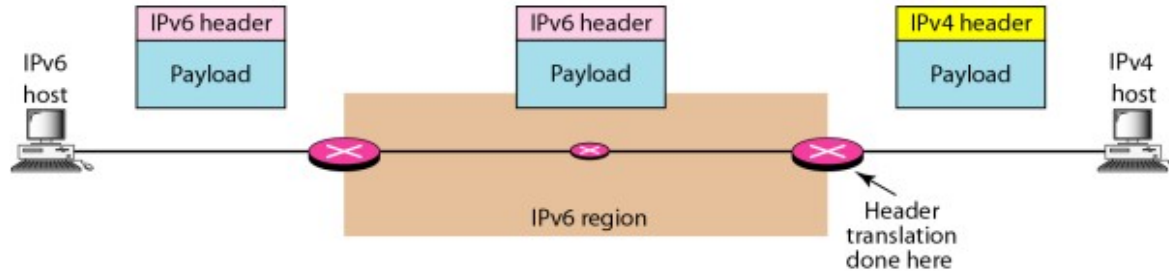
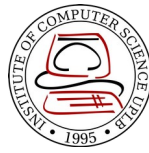
Tunneling Strategy



The IPv6 packet is encapsulated in an IPv4 packet when it enters the region and it leaves the capsule when it exits the region



Header Translation



Header Translation Procedure

1. The IPv6 mapped address is changed to an IPv4 address by extracting the rightmost 32 bits.
2. The value of the IPv6 priority field is discarded.
3. The type of service field in IPv4 is set to zero.
4. The checksum for IPv4 is calculated and inserted in the corresponding field.
5. The IPv6 flow label is ignored.
6. Compatible extension headers are converted to options and inserted in the IPv4 header. Some may have to be dropped.
7. The length of IPv4 header is calculated and inserted into the corresponding field.
8. The total length of the IPv4 packet is calculated and inserted in the corresponding field.

References



- Forouzan, B.A. 2013. Data Communications and Networking, 5th Ed. McGraw-Hill, New York.